Exp No:5 Experiment to detect outliers in a given data set.

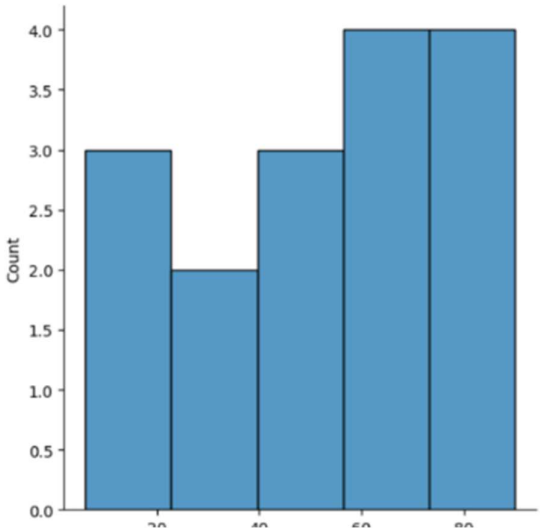
CODE

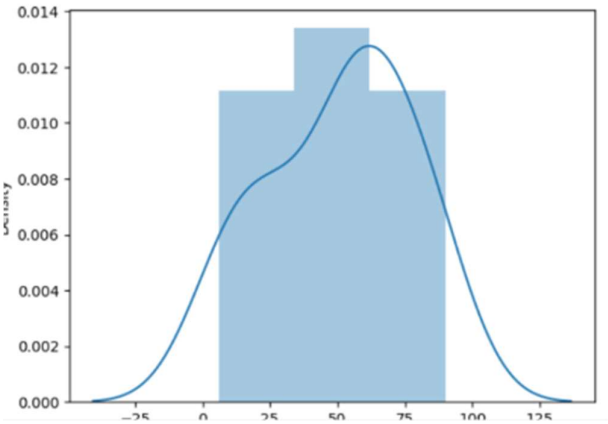
```
import numpy as np
array=np.random.randint(1,100,16)
array.mean()
np.percentile(array,25)
np.percentile(array,50)
np.percentile(array,75)
np.percentile(array,100)
def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)
lr,ur
import seaborn as sns
%matplotlib inline
sns.displot(array)
sns.distplot(array)
new_array=array[(array>lr) & (array<ur)]
```

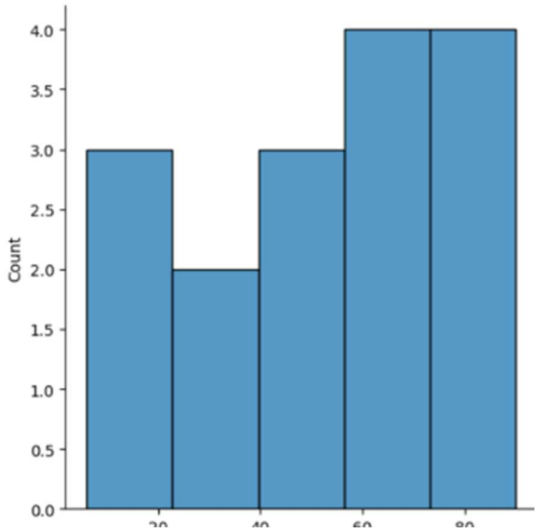
```
sns.displot(new_array)
lr1,ur1=outDetection(new_array)
lr1,ur1
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

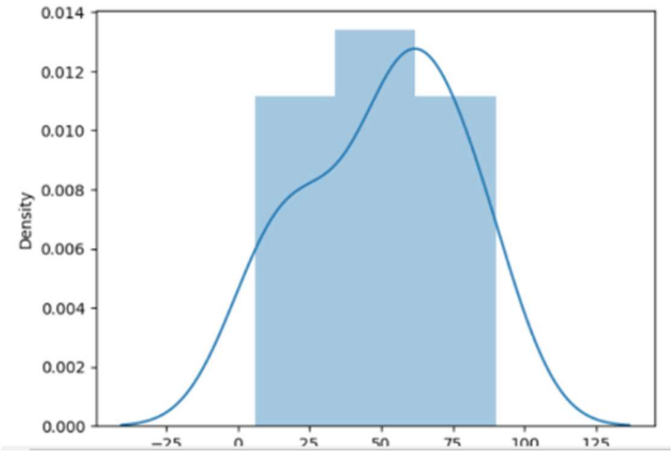
OUTPUT

```
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
50.5
26.0
56.0
69.0
(-38.5, 133.5)
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
```









RESULT hence the outliers detected in a given data set.